

✓ Q1. Create a Hospital Database

```
import sqlite3

conn = sqlite3.connect('hospital.db')
cur = conn.cursor()
```

✓ Employees (Management Team)

```
cur.execute("""
CREATE TABLE Employees (
    emp_id INTEGER PRIMARY KEY,
    emp_name TEXT,
    role TEXT,
    dept_id INTEGER,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
)
""")
```

```
<sqlite3.Cursor at 0x7e185ac65d40>
```

✓ Departments

```
cur.execute("""
CREATE TABLE Departments (
    dept_id INTEGER PRIMARY KEY,
    dept_name TEXT
)
""")
```

```
<sqlite3.Cursor at 0x7e185ac65d40>
```

✓ Doctors and Nurses Records

```
cur.execute("""
CREATE TABLE Doctors_Nurses (
    staff_id INTEGER PRIMARY KEY,
    staff_name TEXT,
    designation TEXT,
    dept_id INTEGER,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
)
""")
```

```
<sqlite3.Cursor at 0x7e185ac65d40>
```

✓ Operatable Rooms / Wards

```
cur.execute("""
CREATE TABLE Rooms_Wards (
    room_id INTEGER PRIMARY KEY,
    room_type TEXT,
    dept_id INTEGER,
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)
)
""")
```

<sqlite3.Cursor at 0x7e185ac65d40>

✓ Patient Records

```
cur.execute("""
CREATE TABLE Patients (
    patient_id INTEGER PRIMARY KEY,
    patient_name TEXT,
    assigned_staff INTEGER,
    room_id INTEGER,
    FOREIGN KEY (assigned_staff) REFERENCES Doctors_Nurses(staff_id),
    FOREIGN KEY (room_id) REFERENCES Rooms_Wards(room_id)
)
""")
```

<sqlite3.Cursor at 0x7e185ac65d40>

✓ Inserting Records into Tables

```
cur.executescript("""
INSERT INTO Departments VALUES
(1, 'Administration'),
(2, 'Cardiology'),
(3, 'Neurology'),
(4, 'Emergency');

INSERT INTO Employees VALUES
(1, 'Aayush Acharya', 'Hospital Director', 1),
(2, 'Aarjan Mani Kandel', 'Operations Manager', 1),
(3, 'Swornim Dangol', 'HR Manager', 1);

INSERT INTO Doctors_Nurses VALUES
(1, 'Nirajan Kumar Yadav', 'Cardiologist', 2),
(2, 'Shiksha Karki', 'Neurologist', 3),
(3, 'Subu Nepal', 'Emergency Nurse', 4),
(4, 'Saramsha Aryal', 'Staff Nurse', 2);

INSERT INTO Rooms_Wards VALUES
(1, 'ICU', 2),
```

```
(2, 'General Ward', 3),
(3, 'Emergency Ward', 4);

INSERT INTO Patients VALUES
(1, 'Nilima Shrestha', 1, 1),
(2, 'Rohit Adhikari', 2, 2),
(3, 'Pratik Bhandari', NULL, 3);
""")

conn.commit()
```

✓ Select Query using LEFT JOIN and Aliases

```
cur.execute("""
SELECT
    p.patient_name AS Patient,
    d.staff_name AS Assigned_Staff,
    d.designation AS Role,
    r.room_type AS Room,
    dept.dept_name AS Department
FROM Patients p
LEFT JOIN Doctors_Nurses d ON p.assigned_staff = d.staff_id
LEFT JOIN Rooms_Wards r ON p.room_id = r.room_id
LEFT JOIN Departments dept ON r.dept_id = dept.dept_id
""")
```

```
for row in cur.fetchall():
    print(row)
```

```
('Nilima Shrestha', 'Nirajan Kumar Yadav', 'Cardiologist', 'ICU', 'Cardiology')
('Rohit Adhikari', 'Shiksha Karki', 'Neurologist', 'General Ward', 'Neurology')
('Pratik Bhandari', None, None, 'Emergency Ward', 'Emergency')
```

```
conn.close()
```

